

Optimal Event Trading

Introduction

Event driven hedge funds rely on forming views on catalysts to generate PnL. In the case of earnings, a PM can generate 25% to 50% of their PnL from these bets. Thus, it is important to develop a framework for positioning and trading around events.

Market Impact Model

As the foundational market-impact model, we present the framework of Almgren and Chriss [1]. An investor holding X shares must liquidate the full position by a fixed horizon T . Liquidating quickly concentrates trading into a short window and pushes execution prices away from the pre-trade price: **market impact**. Liquidating slowly avoids that cost but leaves the position exposed to the market's random drift for longer: **timing risk**. Almgren and Chriss cast this trade-off as a mean-variance optimization over the liquidation schedule.

Discretize the horizon into N steps of length $\tau = T/N$. Let x_k denote the shares still held after step k , with $x_0 = X$ and $x_N = 0$, and let $n_k = x_{k-1} - x_k$ be the shares traded in step k . Three parameters drive the cost: permanent impact γ (the lasting price shift per share traded, which the market never unwinds), temporary impact η (the transient price concession demanded for trading at a given rate, which decays once trading stops), and volatility σ (the size of the position's random price moves per unit time).

Permanent impact contributes $\frac{1}{2}\gamma X^2$ to expected cost: each share traded shifts the price by a further increment of γ , so the average share pays half the full depression γX , giving a total of $X \cdot \frac{1}{2}\gamma X = \frac{1}{2}\gamma X^2$. This is fixed by the total size X and unaffected by the schedule, since every admissible schedule liquidates the same X . Temporary impact contributes $(\eta/\tau) \sum_{k=1}^N n_k^2$, because the cost of a step is quadratic in its trading rate, spreading the same shares over more time reduces it. The variance of total cost, driven by how many shares remain exposed and for how long, is $\sigma^2 \sum_{k=1}^N \tau x_k^2$. Almgren and Chriss minimize

$$\mathbb{E}(\text{cost}) + \lambda \mathbb{V}(\text{cost})$$

over the schedule $x_1, \dots, x_{N-1}$, where $\lambda \geq 0$ is the trader's risk aversion. The full derivation, taken to its continuous-time limit, is given in the Appendix.

The optimal schedule is

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad \kappa = \sqrt{\lambda \sigma^2 / \eta}.$$

The single parameter κ governs the shape of the liquidation curve. As $\lambda \rightarrow 0$ (risk-neutral), $\kappa \rightarrow 0$ and the schedule becomes linear: a constant liquidation rate, i.e. minimum-impact, TWAP-like execution. As λ grows, κ grows and the curve front-loads: more of the position is sold early, trading higher impact cost for lower exposure to timing risk. The quantity $1/\kappa$ sets the characteristic time scale (the "half-life") over which the position is unwound.

Temporary price impact is taken to be linear in the trading rate: executing at rate $\dot{x}(t)$ displaces the execution price by an amount proportional to $\dot{x}(t)$. Because the cost accrued over an instant is the product of this displacement and the amount traded, it is quadratic in the rate, consistent with the $\eta n_k^2 / \tau$ term above.

At calendar time t you trade at a dollar rate $\dot{x}(t)$ against a market dollar volume rate $v(t)$; the participation rate is

$$\pi(t) = \dot{x} \frac{t}{v(t)}.$$

Heuristic Model

We begin by developing a minimal model that isolates the essential trade-off governing event-driven position sizing. The trade is reduced to two competing terms: the expected profit accruing from the position held on the event date, $\alpha(t)x(t)$, and the transaction cost incurred in establishing that position. Maximizing their difference yields a closed-form expression for the optimal size. Despite its simplicity, the model recovers the standard heuristics used by practitioners and provides quantitative guidance that we refine in the more general treatment that follows.

References

- [1] R. Almgren and N. Chriss, “Optimal execution of portfolio transactions,” *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.

Appendix

Notation

symbol	meaning	units
$v(t)$	market's instantaneous dollar volume rate: the rate at which the tape prints, at calendar time t	\$/day
$\tau(t) = \int_0^t v(s) ds$	cumulative market dollars traded since $t = 0$	\$
V	a fixed reference daily volume: ADV, used only to rescale	\$/day
$\dot{x}(t)$	your dollar trading rate	\$/day
$x(t)$	position, in dollars	\$
$\alpha(t)$	expected return over the event	dimensionless
$\pi(t) = \dot{x}(t)/v(t)$	participation rate: your trading rate as a fraction of market volume	dimensionless

Almgren-Chriss derivation

Following [1], the discrete-time objective introduced in the Market Impact Model section is

$$\min_{x_1, \dots, x_{N-1}} \underbrace{\frac{1}{2}\gamma X^2 + \frac{\eta}{\tau} \sum_{k=1}^N n_k^2}_{\mathbb{E}[\text{cost}]} + \underbrace{\lambda \sigma^2 \sum_{k=1}^N \tau x_k^2}_{\mathbb{V}[\text{cost}]}.$$

The permanent-impact term $\frac{1}{2}\gamma X^2$ does not depend on the schedule $x_1, \dots, x_{N-1}$ and drops out of the minimization; only the temporary impact and variance terms shape the optimal path.

Continuous limit. Taking $N \rightarrow \infty$ with $\tau = T/N \rightarrow 0$, write $x(t)$ for the holdings path and $\dot{x}(t) = \lim_{\tau \rightarrow 0} (-n_k/\tau)$ for the (negative) trading rate. The sums become integrals and the objective reduces to

$$\min_{x(\cdot)} \int_0^T [\eta \dot{x}(t)^2 + \lambda \sigma^2 x(t)^2] dt, \quad x(0) = X, \quad x(T) = 0.$$

Euler-Lagrange. This is a standard calculus-of-variations problem with Lagrangian $L(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$. The Euler-Lagrange condition $\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$ gives

$$2\eta \ddot{x}(t) = 2\lambda \sigma^2 x(t) \quad \Rightarrow \quad \ddot{x}(t) = \kappa^2 x(t), \quad \kappa^2 = \lambda \frac{\sigma^2}{\eta}.$$

Solving the ODE. The general solution of $\ddot{x} = \kappa^2 x$ is $x(t) = A \sinh(\kappa t) + B \cosh(\kappa t)$. Imposing $x(T) = 0$ and $x(0) = X$ and solving for A, B collapses this to the single closed form

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}.$$

Risk-neutral limit. As $\kappa \rightarrow 0$ the expression is a 0/0 indeterminate; using $\sinh(z) \approx z$ for small z ,

$$\frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)} \xrightarrow{\kappa \rightarrow 0} \frac{\kappa(T-t)}{\kappa T} = \frac{T-t}{T},$$

so $x(t) = X(1 - t/T)$: constant-rate liquidation, recovering the $\lambda = 0$ (pure minimum-impact) case.